

Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories — spending patterns, customer segments, and subscription behavior to guide strategic business decisions.



Dataset at a Glance

3,900

Total Purchases

18

Data Columns

37

Missing Values

Review Rating column

\$59.76

Avg Purchase

Features span customer demographics (Age, Gender, Location, Subscription), purchase details (Item, Category, Amount, Season), and shopping behavior (Discount, Promo Code, Frequency, Review Rating, Shipping Type).

Data Preparation in Python

→ Load & Explore

Imported via pandas; used `df.info()` and `df.describe()` for structure and statistics.

→ Clean & Standardize

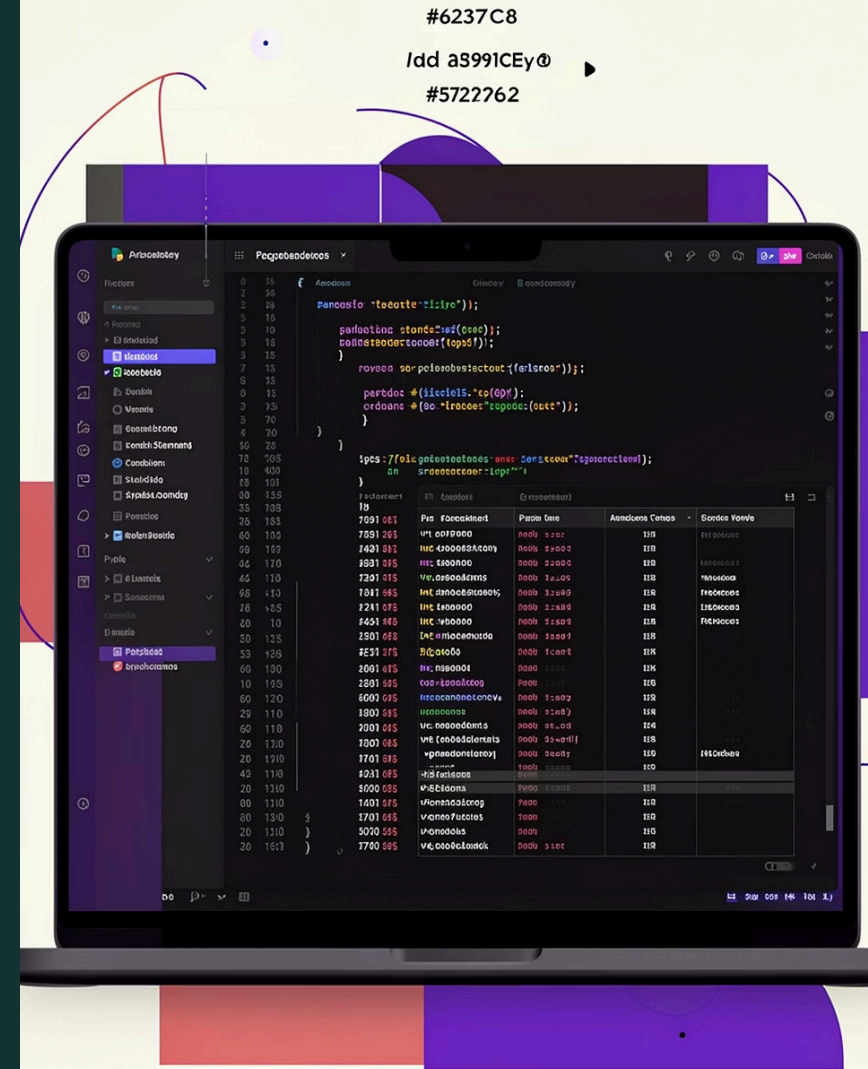
Imputed missing Review Ratings using median by category;
renamed columns to snake_case.

→ Feature Engineering

Created `age_group` (binned ages) and `purchase_frequency_days`; dropped redundant `promo_code_used`.

→ Database Integration

Loaded cleaned DataFrame into PostgreSQL for SQL analysis.



Revenue by Gender & Subscription



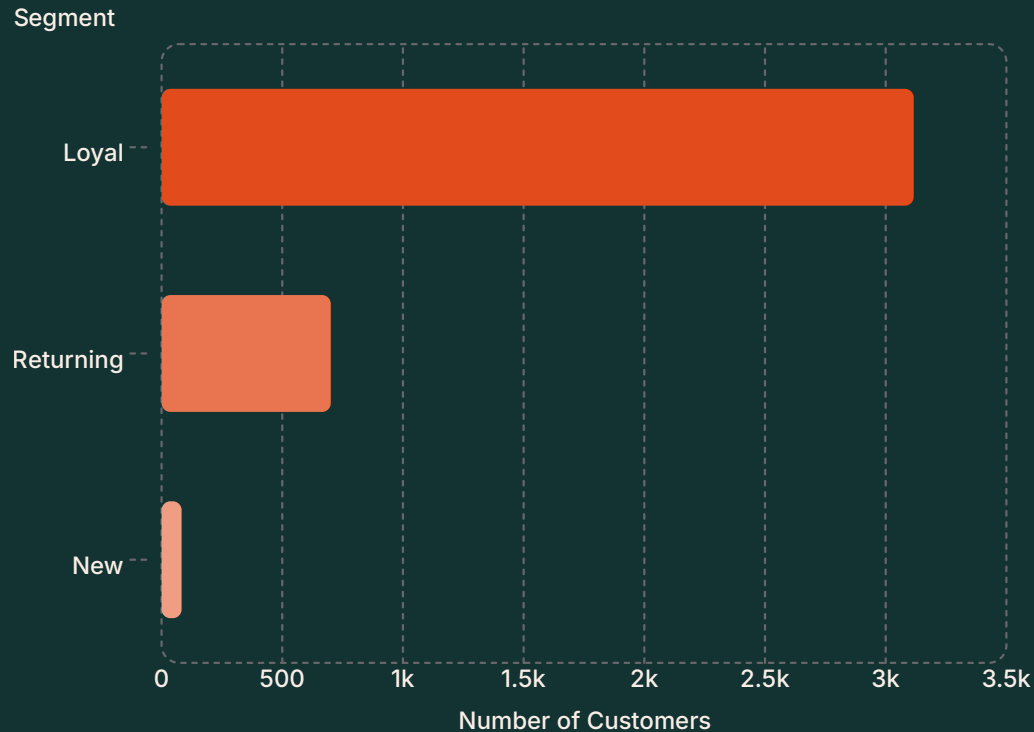
Key Findings

Male customers generated **2.1× more revenue** than female customers (\$157,890 vs \$75,191), despite females representing a notable share of buyers.

Subscribers (1,053 customers) averaged \$59.49 spend with \$62,645 total revenue. Non-subscribers (2,847 customers) averaged \$59.87 spend with \$170,436 total revenue.

Subscription status shows minimal impact on per-customer spend.

Customer Segmentation



Segment Breakdown

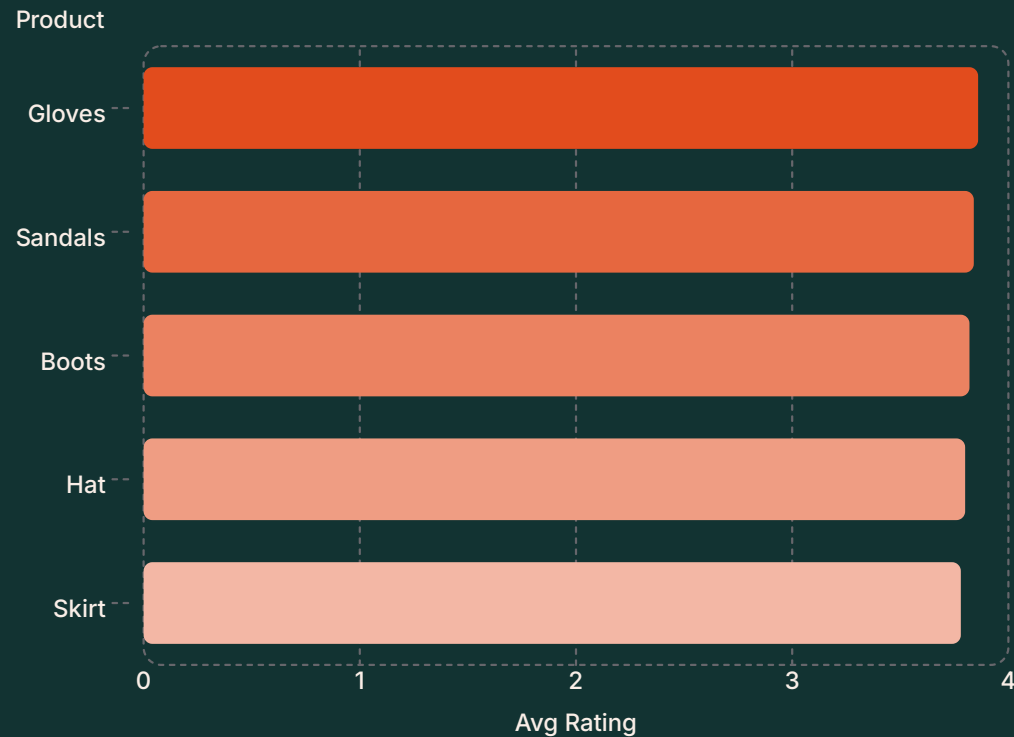
Customers classified by purchase history into three segments:

- **Loyal (3,116):** The dominant segment – strong retention base
- **Returning (701):** Moderate repeat buyers with growth potential
- **New (83):** Small but critical for acquisition strategy

Of customers with **>5 purchases**, 958 are subscribers vs 2,518 non-subscribers – repeat buyers are not yet strongly converting to subscriptions.

Top Products & Categories

Top 5 Products by Review Rating



Top Products by Category

Clothing

Blouse (171), Pants (171),
Shirt (169)

Footwear

Sandals (160), Shoes
(150), Sneakers (145)

Outerwear

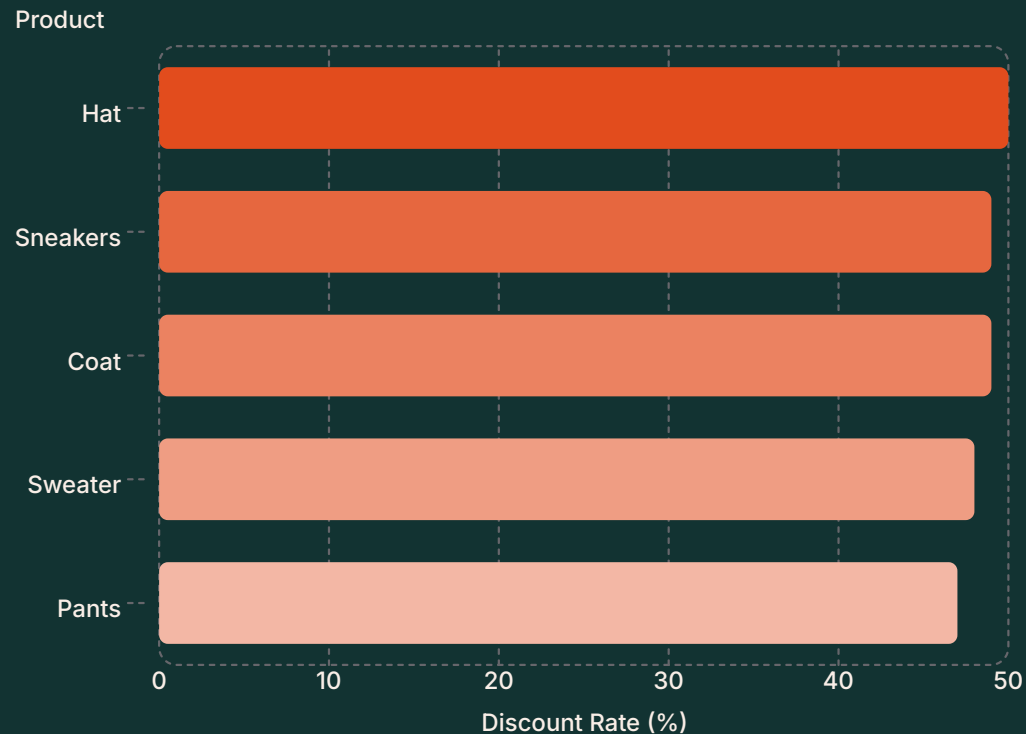
Jacket (163), Coat (161)

Accessories

Jewelry (171),
Sunglasses (161), Belt
(161)

Discount Behavior & Shipping

Products Most Discounted



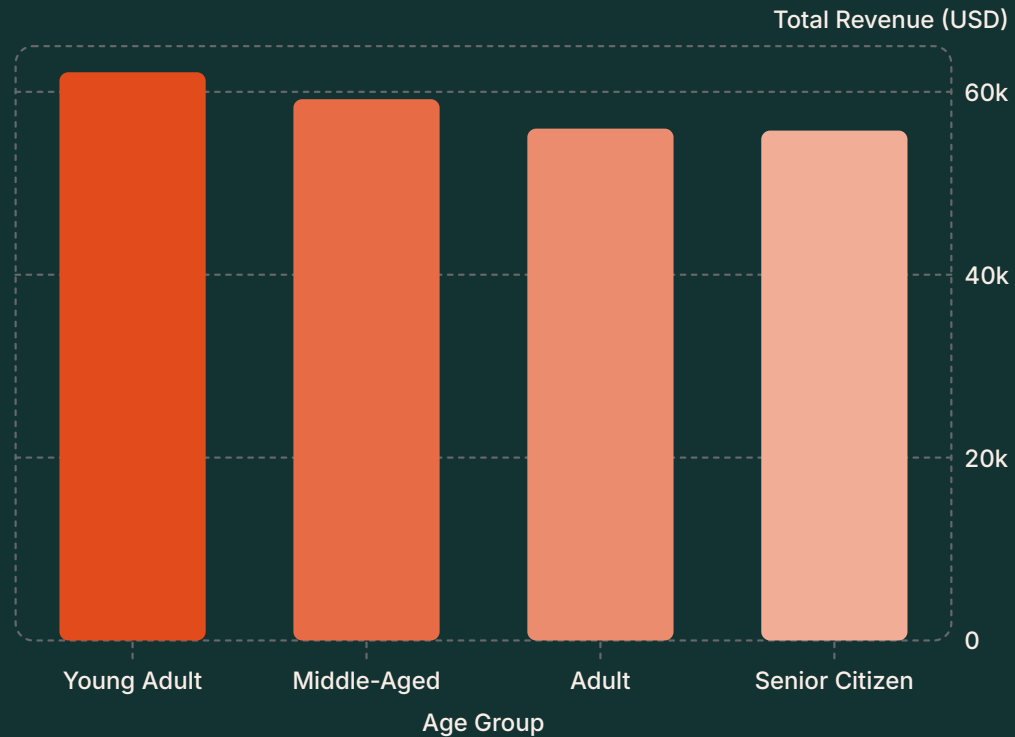
Shipping & Discount Insights

Express shipping averages \$60.48 vs Standard at \$58.46 — a modest premium.

839 customers used discounts yet spent above the average purchase amount, indicating discount users are not solely bargain hunters.

Hats lead discount dependency at **50%** of purchases discounted, followed closely by Sneakers and Coat at 49%.

Revenue by Age Group



Age Group Insights

Young Adults lead revenue contribution at **\$62,143**, with a gradual decline across older groups. The spread between top and bottom groups is relatively narrow (-\$6,380), suggesting broad purchasing power across all age segments.

Average customer age is **44 years** (range 18–70), with 25th percentile at 31 and 75th percentile at 57.

Power BI Dashboard

3.9K Customers

Total transactions tracked

3.75 Avg Rating

Overall review score

\$59.76 Avg Purchase

Mean spend per transaction

Interactive filters include Subscription Status, Gender, Category, and Shipping Type. Visuals cover Revenue & Sales by Category, Subscription split (27% Yes / 73% No), and breakdowns by Age Group.

Business Recommendations



Customer Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment.



Review Discount Policy

Balance sales boosts with margin control.



Product Positioning

Highlight top-rated and best-selling products in campaigns.



Targeted Marketing

Focus efforts on high-revenue age groups and express-shipping users.



Boost Subscriptions

Promote exclusive benefits for subscribers.